RIPHAH INTERNATIONAL UNIVERSITY ISLAMABAD

Faculty of Social Sciences & Humanities

Department of Computer Science

Final Term Examination Spring 2026 Semester

Course Name:	Object-Oriented Programming	**Course Code:**	CS201
Program:	BS (Computer Science)	**Section:**	B
Semester:	Final	**Date of Examination**	01-06-2026
Teacher's Name	Muhammad Usman	**Total Marks**	40
Exam Duration	(10:00 am – 12:00 pm)		

INSTRUCTIONS

- Write your name and SAP ID on all answer sheets.
- Write page number on all sheets.
- Please be analytical and specific in answering questions.
- Switch off your Mobile Phone.
- Attempt all the following questions critically and creatively.

Question 1: **[12 Marks]**

Answer the following short scenario-based questions:

(a) What is the significance of the 'super' keyword in inheritance? Illustrate with a scenario where omitting it causes incorrect behaviour. [3 Marks]

(b) A software company follows the principle of 'hiding internal implementation details.' Which OOP pillar does this represent and how does it benefit large-scale software development? [3 Marks]

(c) Compare compile-time polymorphism and runtime polymorphism. In which situations would each be preferred? Provide one example scenario for each. [3 Marks]

(d) Why are interfaces preferred over abstract classes in some designs? Explain using a scenario involving multiple inheritance. [3 Marks]

Question 2: **[08 Marks]**

(a) Identify and correct ALL errors in the following code:

```
interface Printable {
    void print();
}

abstract class Document implements Printable {
    String title;
    Document(String t) { title = t; }
}
```

```
class Report extends Document {
    Report(String t) { super(t); }

    void print() {
        System.out.println("Printing: " + title);
    }
}

class Main {
    public static void main(String[] args) {
        Printable p = new Document("Annual Report"); // cannot instantiate abstract
class
        p.print();
    }
}
```

(b) Analyze the following code and write the exact output:

```
class Animal {
    String name;
    Animal(String n) { this.name = n; }
    void sound() { System.out.println(name + " makes a sound"); }
}

class Dog extends Animal {
    Dog(String n) { super(n); }
    void sound() { System.out.println(name + " barks"); }
}

class Main {
    public static void main(String[] args) {
        Animal a = new Dog("Rex");
        a.sound();
        Animal b = new Animal("Generic");
        b.sound();
    }
}
```

Question 3: [20 Marks]

Design a Hospital Patient Management System in Java using the following OOP concepts:

1. Inheritance: Create a base class Person and derive Patient and Doctor from it.
2. Encapsulation: Use private attributes for sensitive data (e.g., medical records) with controlled access.
3. Polymorphism: Override a getDetails() method to display appropriate info for Patient and Doctor.
4. Abstraction: Define an abstract class or interface HospitalEntity with abstract methods.
5. File Handling: Write patient appointment details to a file and read them back.
6. Exception Handling: Handle invalid patient ID, file errors, and negative age input using try-catch.

Note: The system should support registering a patient, booking an appointment (stored in file), and displaying records.